

Jump Process

(2. Option Pricing under MJD)

Adding jump to Stock price dynamics

We add the jump component to stock price dynamics as follows

$(0, 0.75, 1, 1.25, \dots)$ → $\frac{dS_t}{S_{t-}} = \mu dt + \sigma dW_t + \text{jump}$

$ds = s \cdot \text{jump}$

$\text{jump} = \begin{cases} \text{Arrival,} \\ \text{Size,} \end{cases} \quad \begin{matrix} \text{Poisson} \\ \log - \text{Normal} \end{matrix} \quad (\lambda)$

Note that in this formulation the jump is a multiplicative component that gets multiplied with S_{t-} . If a jump happened, we could link the stock price before and after the jump,

$S_t = S_{t-} Y_t \Rightarrow \frac{dS_t}{S_{t-}} = Y_t - 1$

$ds/s = Y_t - 1$

We can therefore write the dynamics as

$\frac{dS_t}{S_{t-}} = \mu dt + \sigma dW_t + (Y_t - 1) dN_t$

$ds/s = [Y_t - 1]$

Adding jump to Stock price dynamics

The addition of jump introduces some drift. Let's calculate the drift.

$$\frac{dS_t}{S_{t-}} = \underbrace{\mu dt}_{\text{drift} \neq 0} + \sigma dW_t + \underbrace{(Y_t - 1)dN_t}_{\text{jump}}$$

$$dN_t \sim \text{Poi}(\lambda dt)$$

$$\mathbb{E}[(Y_t - 1)dN_t] = \underbrace{\mathbb{E}[Y_t - 1]}_k \mathbb{E}[dN_t] = k\lambda dt$$

We can refine our SDE to ensure no additional drift is included as part of jump by using a compensation term.

$$\frac{dS_t}{S_{t-}} = \mu dt + \sigma dW_t + (Y_t - 1)dN_t - k\lambda dt$$

In the interval dt the net multiplicative jump can be written as the product of all the multiplicative jumps.

$$Y_1 Y_2 \dots Y_{dN_t}$$

$$S_t = (S_{t-})^{Y_1 Y_2 Y_3}$$

$$\frac{dS_t}{S_{t-}} = (\mu - k\lambda)dt + \sigma dW_t + \left(\prod_{j=1}^{dN_t} Y_j - 1 \right)$$

$$(S_{t-})^1 = S_{t-}$$

Ito's lemma for $f = \ln S$

Let's derive the dynamics of $f(S) = \log S$

$$\rightarrow \underbrace{dS_t}_{\text{circled}} = \underbrace{(S_{t-})}_{\text{circled}} (\underbrace{\mu - k\lambda}_{\text{circled}}) dt + \underbrace{(S_{t-})}_{\text{circled}} \sigma dW_t + \underbrace{(S_{t-})}_{\text{circled}} \left(\underbrace{\prod_{j=1}^{dN_t} Y_j - 1}_{\text{circled}} \right)$$

$f(s) \rightarrow f(+, s)$
 $(S_{t-}, Y_1, Y_2, \dots, Y_{dN}) = S$

$$\underbrace{df(S_t)}_{\text{circled}} = \underbrace{\left(\frac{\partial f}{\partial S} \right)}_{\text{circled}} dS + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} (dS)^2 + \underbrace{f \left(S \prod_{j=1}^{dN_t} Y_j \right) - f(S)}_{\text{circled}} \rightarrow \text{circled } f$$

$$\underbrace{d \log S}_{\text{circled}} = \underbrace{\left(\frac{1}{S} \right)}_{\text{circled}} dS - \frac{1}{2} \left(\frac{dS}{S} \right)^2 + \underbrace{\log \left(S \prod_{j=1}^{dN_t} Y_j \right) - \log S}_{\text{circled}} = \log \left(\frac{\prod_{j=1}^{dN_t} Y_j}{1} \right)$$

$= \sum_{j=1}^{dN_t} \log Y_j$

$$\boxed{d(\log S) = \left(\mu - k\lambda - \frac{1}{2} \sigma^2 \right) dt + \sigma dW + \sum_{j=1}^{dN_t} \log Y_j}$$

$$S_t = (S_0 Y_1 Y_2 Y_3 \dots)$$

$$\ln S_t = \boxed{\ln S_0} + \ln Y_1 + \ln Y_2 + \dots$$

Solution – (I)

$$d \log S = \left(\mu - k\lambda - \frac{1}{2}\sigma^2 \right) dt + \sigma dW + \sum_{j=1}^{dN_t} \log Y_j$$

Handwritten notes: 0 to t, sigma W_t, dN_t, N_t

$$\log S_t - \log S_0 = \left(\mu - k\lambda - \frac{1}{2}\sigma^2 \right) t + \underbrace{N(\mu_y, \sigma^2 t + n\sigma_y^2)}_{\text{a normal random}}$$

$$\log S_t - \log S_0 = \left(\mu - k\lambda - \frac{1}{2}\sigma^2 \right) t + n\mu_y + \sqrt{\sigma^2 + \frac{n\sigma_y^2}{t}} W_t$$

Handwritten notes: z\sqrt{t}

$$\log S_t - \log S_0 = \left(\mu - k\lambda - \frac{1}{2}\sigma_n^2 + \frac{n\sigma_y^2}{2t} \right) t + n\mu_y + \sigma_n W_t$$

Handwritten notes: (-1/2\sigma_n^2) + 1/2\sigma_n^2

$$\log S_t - \log S_0 = \left(\mu - \frac{1}{2}\sigma_n^2 \right) t + \left(n\mu_y - k\lambda t + \frac{n\sigma_y^2}{2} \right) + \sigma_n W_t$$

*Handwritten notes: (sigma^2 + n\sigma_y^2/t) * (sqrt{t}) * (+) (sigma^2 + n\sigma_y^2/t)*

Assume $(\log Y_j) \sim N(\mu_y, \sigma_y^2)$, Prove $k = e^{\mu_y + \frac{1}{2}\sigma_y^2} - 1$

Handwritten notes: k = E[X-1] = E[Y]-1 = e^{\mu_y + 1/2\sigma_y^2} - 1

Assume realization of $N_t = n$

Define $\sigma_n = \sqrt{\sigma^2 + \frac{n\sigma_y^2}{t}}$

$$\sigma W_t + N(n\mu_y, n\sigma_y^2)$$

$$N(0, \sigma^2 t) + N(n\mu_y, n\sigma_y^2)$$

$$N(n\mu_y, \sigma^2 t + n\sigma_y^2)$$

Handwritten notes: n\mu_y + N(0, \sigma^2 t + n\sigma_y^2)

Solution – (II)

$$\log S_t - \log S_0 = \left(\mu - \frac{1}{2} \sigma_n^2 \right) t + \left(n\mu_y - k\lambda t + \frac{n\sigma_y^2}{2} \right) + \sigma_n W_t$$

$\log S_t - \log S_0 = (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$
 $\rightarrow S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$

$$\log S_t - \log S_0 = \left(n\mu_y - k\lambda t + \frac{n\sigma_y^2}{2} \right) + \left(\mu - \frac{1}{2} \sigma_n^2 \right) t + \sigma_n W_t$$

Define $S_0^{(n)} = S_0 e^{(n\mu_y - k\lambda t + \frac{n\sigma_y^2}{2})}$

$$S_t = S_0^{(n)} \exp \left(\left(\mu - \frac{1}{2} \sigma_n^2 \right) t + \sigma_n W_t \right)$$

$S_t = S_0 e^{(n\mu_y - k\lambda t + \frac{n\sigma_y^2}{2})} e^{(\mu - \frac{1}{2}\sigma_n^2)t + \sigma_n W_t}$

Under risk neutral dynamics, we can rewrite as

$$S_t = S_0^{(n)} \exp \left(\left(r - \frac{1}{2} \sigma_n^2 \right) t + \sigma_n W_t \right)$$

Adjusted spot Adjusted vol

$(\mu, \sigma) \rightarrow (r, \sigma)$
 $\mathbb{E}^Q(S_T) = S_0 e^{rT}$

This is in the familiar GBM solution form where the initial stock price and the volatility are modified. We can get the closed form Black-Scholes formula for European option price. But note that the price is conditional on $N_t = n$. We can obtain the Option price by using tower property

$(0, T)$

$$C(S_0, T) = \sum_{n=0}^{\infty} C(S_0^{(n)}, T | N_t = n) \mathbb{P}(N_t = n)$$

$$\mathbb{E}^Q(S_T) = S_0 e^{rT}$$

Terminal density S_T

Under Black-Scholes the terminal density of stock price is lognormal. Under MJD, we can calculate the terminal density using the infinite sum as the stock price is conditionally lognormal (or log stock price is conditionally normal)

$$\log S_T = \log S_0 + \left(n\mu_y - k\lambda T + \frac{n\sigma_y^2}{2} \right) + \left(r - \frac{1}{2}\sigma_n^2 \right) T + \sigma_n W_t$$

$$\log S_T \sim N(\mu_n, v_n)$$

$$\mu_n = \log S_0 + \left(n\mu_y - k\lambda T + \frac{n\sigma_y^2}{2} \right) + \left(r - \frac{1}{2}\sigma_n^2 \right) T, \quad v_n = \sigma_n^2 T$$

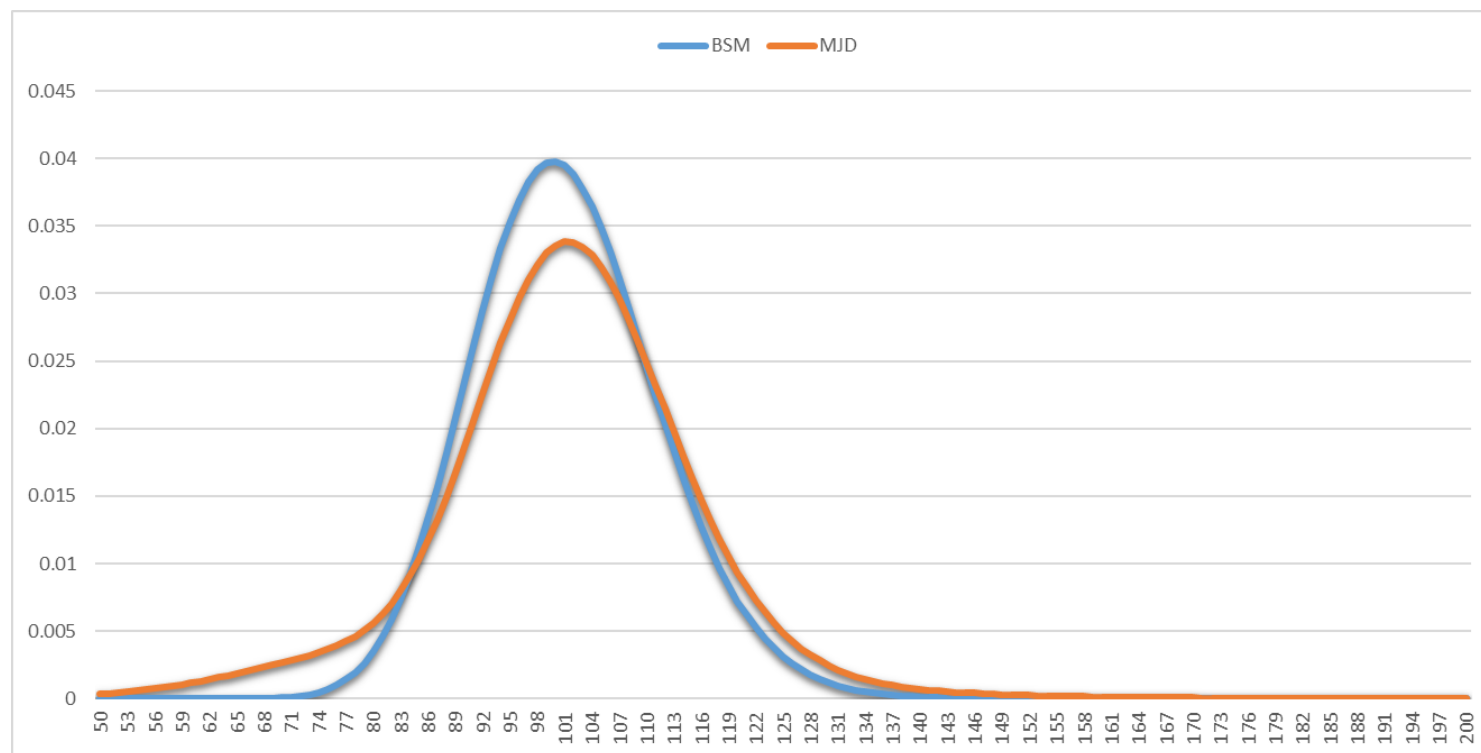
$$S_T \sim \text{lognormal}(\mu_n, v_n)$$

$$\ln S_T - \ln S_0 = \left(\mu - \frac{1}{2}\sigma^2 \right) T + \sigma W_T$$

$$\ln S_T \sim N \left(\ln S_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) T, \sigma^2 T \right)$$

$$S_T \sim \text{lognormal}(\mu, v)$$

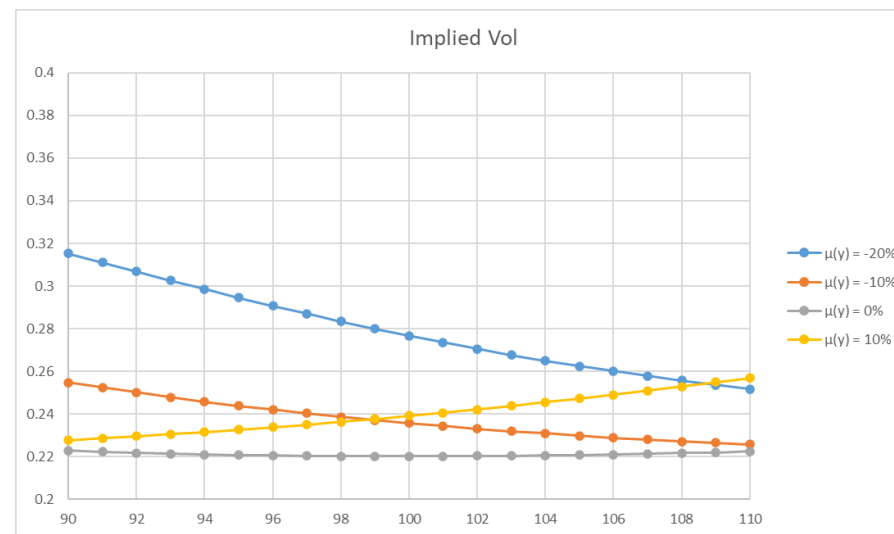
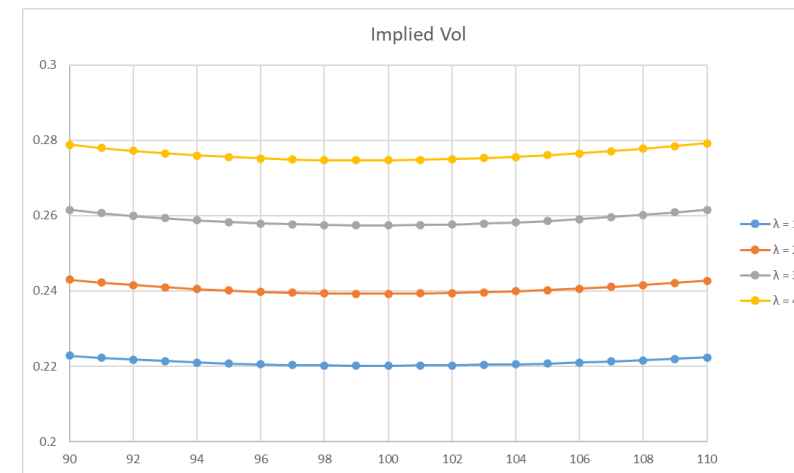
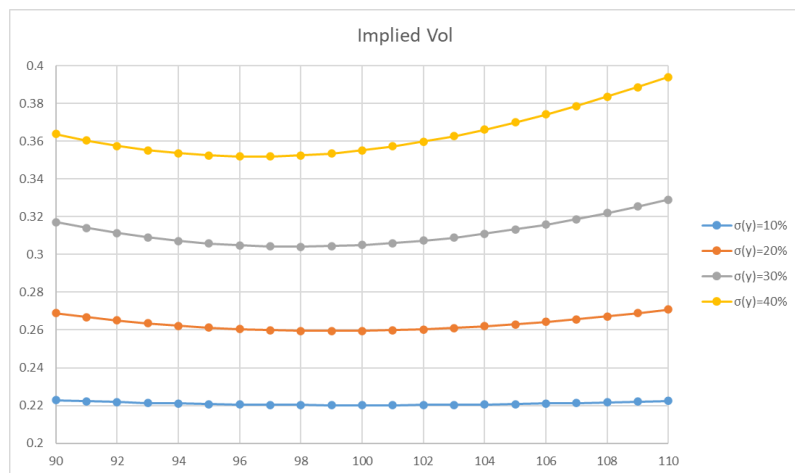
Terminal Density (BSM vs MJD)



σ	20%
$\mu(y)$	-10%
$\sigma(y)$	20%
λ	1

MJD shows fatter tails

Implied Volatility



Exercise



1. Calibrate MJD to the option chain of your favorite stock. Plot the observed and fitted implied volatility.
2. Simulate terminal stock price under MJD. Compare the Monte Carlo price with infinite Poisson weighted Black-Scholes formula.

